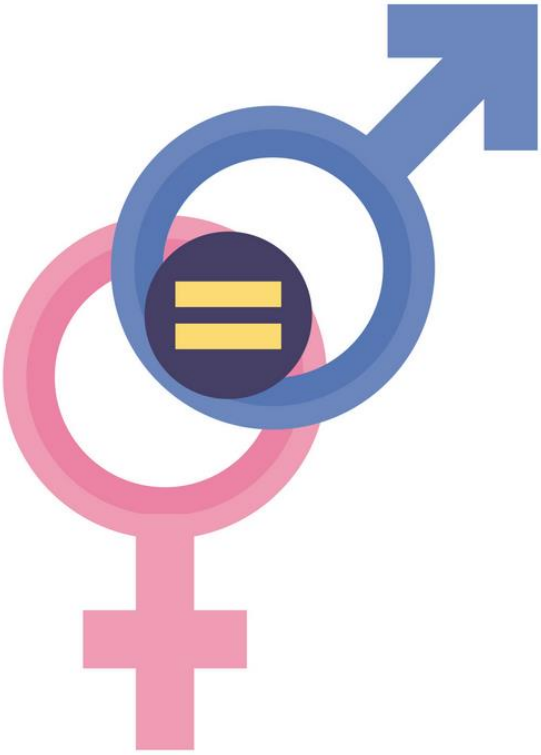
Achieve gender equality and empower all women and girls

[1]



Home Assignment

DS819: Data Driven Applications

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[1] Photo by TRRAIN

Introduction and Problem formulation

This project explores UN Sustainable Development Goal 5 on Gender Equality and focuses on Target 5.3: "*Eliminate harmful practices such as child, early and forced marriage and female genital mutilation*"[2] as a theme for the theme website.

The topic of this project has been chosen to raise awareness and to support the millions of women and girls who have faced the consequences of these harmful social and cultural practices. Despite international commitments and tremendous progress towards gender equality in recent decades, inequalities persist worldwide whether its in educational attainment and skills, wages at work, health, gender-based violence and other policy areas.

The aim of the project is therefore to:

"*Raise awareness on forced marriage and female genital mutilation and suggest specific actions that people can take to achieve gender equality and empower all women and girls*"

And in addition to that build a data-driven web application with publicly available data. The application will then demonstrate how data can be transformed into meaningful information for users while raising awareness on forced marriage and FGM.

Research questions

The following research questions will be discussed and done to reach the goal of the project.

Research Question 1:
How can data from multiple web APIs be combined to communicate the global extent of child marriage and FGM?

Research Question 2:
How can data visualizations and narrative structure improve understanding of gender-based harmful practices?

Research Question 3:
How can data communication principles support ethical and clear presentation of sensitive topics?

Narrative Structure

In the beginning of the website, the user is indirectly introduced to goal 5.3. There is a slideshow giving information on EmpowerHer and Gender Equality. Briefly explaining why it's important.

[2] https://sdgs.un.org/goals/goal5#targets_and_indicators

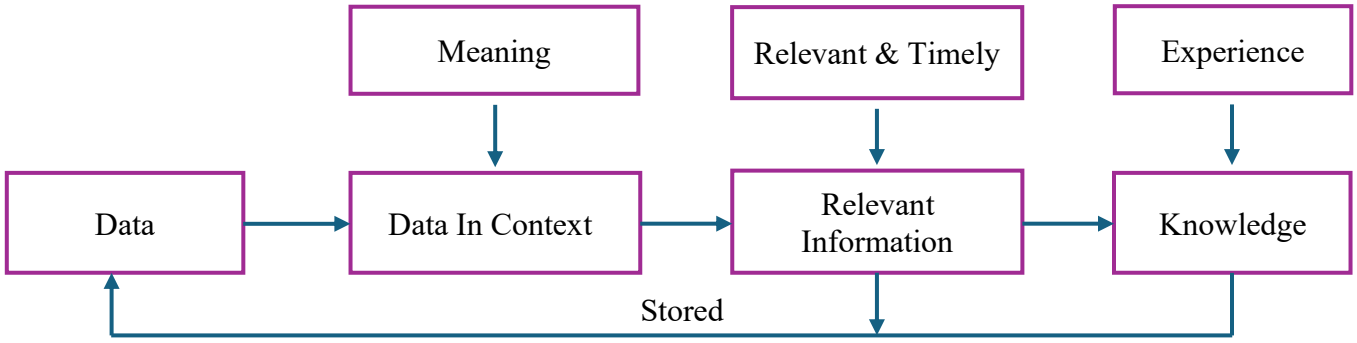
From there we reach the middle, which gives the user a lot more information on FGM and forced marriage. Through data, statistics, media etc. to show how prevalent these practices happen.

In the end a video is presented to give more information on the importance of global action and worldwide empowerment. It's possible for the user to scroll and find books, data and explanations on the topics. Lastly, they can reach out to the EmpowerHer for help, questions etc.

Theoretical background

What is the Data-Information-Knowledge Cycle?

The Data-Information-Knowledge Cycle describes the process, as seen in the model below, through which raw facts transform into meaningful understand.
The first element in the cycle begins with "data", which consist of unprocessed facts without any inherent meaning such as a numbers or text retrieved from a CSV, - and JSON files or web API responses. From here the data gains meaning when it's placed in context within a timeframe and relevance. When the data has become contextualized, relevant, processed and visualized in different forms like tables, charts and infoboxes it becomes information. Information is transformed intro knowledge, when the user of the website interprets it and from here develops an understanding that can inform awareness or decision-making.



The data, in this project, consists of raw facts retrieved from different web APIs and datasets from open access data-repositories, including ? on forced marriage and FGM indicators. Furthermore, from the Open Library Books API to spread educational literature on FGM whereas the NYTimes API gives a journalistic coverage of FGM. To back this data up, additional datasets related to FGM and Forced Marriage prevalence were retrieved and stored locally as CSV and JSON formats. For all this data and information to become knowledge on the website, the user must interpret the visualized data within the narrative structure of it. By mixing contextual media, statistical data and

educational resources the website supports the users in the understanding of implications of harmful practices. (Lecture 1, slide 19 & Dataversity)[3]

Document Object Model

Document Object Model (DOM) is a programming interface for web pages. DOM is used to display and enable a dynamic insertion of data/information from different APIs without reloading. In this project, DOM manipulation is used to create infoboxes and news articles based on API responses. In this project Javascript "Infoboxs.js" select elements from HTML using the function from "infobox.js" line 8-9

```
const content = document.getElementById("content");
if (!content) return;
```

and by creating new elements dynamically using, "infobox.js" line 12,

```
const box = document.createElement("div");
```

These elements are then populated with data parsed from json api responses and put into the website with, "infobox.js" line 12

```
content.appendChild(box);
```

Through DOM the website can insert the data from various APIs into the narrative structure in this way the webpages content is populated programmatically rather than hardcoded in the HTML.

CSV and JSON format

CSV represents data in a tabular with rows and columns representing a variable.
JSON is the dominant format used by web APIs its structured data from key-value pairs and hierarchical objects. JSON data is retrieved from API using JavaScrips "fetch()" and ".json()" functions to display specific fields dynamically on the website.
In the example below the data is retrieved from NY Times giving articles on FGM.

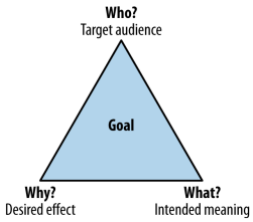
```
async function createInfobox() {
  const response = await fetch(
    "https://openlibrary.org/search.json?q=Female+Genital+Mutilation"
  );
  const data = await response.json();

  const content = document.getElementById("content");
  if (!content) return;
```

"infobox.js" line 2-12

[3] https://www.dataversity.net/articles/the-data-information-knowledge-cycle/

Data Communication Principles



The first 4 principes are Know your goal, Use the right data, Select suitable visualizations and Design for aesthetics

This principle is to ensure a clear goal. It's important to remember the target audience, the intended meaning and desired effect
In other words, data should be shown at a level that is appropriate for the intended users. The aim of this project was to raise awareness on harmful practices done on women and girls, so the audience consists of users interested in gender equality. The data has hopefully been presented in a clear way. The users can be women, families, relatives who have been affected of such practices, people who have an interest in getting more information on it etc.

Use the right data
The right data must be represented to define the message. It's therefore important to ensure that the website gives the desired and related information on goal 5.3, why its relevant, who has been affected and how to help.

Select suitable visualizations
The goal of the website communication was to raise awareness on harmful practices to users interested in gender equality to victims, offenders, people with a general interest etc. to gain more knowledge within the field. This has been done with the help of visualization and text to illustrate the scale and impact of forced marriage and FGM.

For instance, through a table for exact quantities and summary of the information. In this way the users can see how many women and girls who have been affected. Furthermore, we have the line charts to present the data over time. (lecture 1, slide 26-34)

Design for aesthetics

Aesthetics were used to support comprehension rather than just for "decorations". Infoboxes, charts, and media elements were visually in use to reinforce the narrative flow. The aesthetics are here to guide the users through the website by having a beginning, middle and end approach. The different parts of the structure are there to ensure understand and inform on the topic. This is done with the help of different visualization and media elements. The website has a consistent color palette of bright and "girly" colors such as pink and purple. The information is only relevant to the topic to establish cohesion and alignment throughout the page.

API

The New York Times Article Search API was used to retrive journalistic articles related to FGM. A RESTful GET using fetch() function was sent to the Article Search endpoint using relevant query parameters to filter articles by topic and news category. The API returned a JSON response containing metadata for each article. From this response, selected fields such as the headline, abstract, publication date, and article URL were extracted. The content was dynamically inserted into the webpage using DOM manipulation, presenting the articles as cards with links to the original sources. This approach gives information that complements the statistical data presented elsewhere on the website..

The Open Library Books API was used through RESTful GET request with the search term Female Genital mutilation. This API response give the cover images, title and author of “Desert Flower” and “Forced Marriage” which are books related to this topic. The information has been displayed in infoboxes. The idea here is that the user can chose a book on the topic and learn more.

```
box.innerHTML = `
  ${coverHTML}
  <div><strong>Title:</strong> ${book.title ?? "Unknown"}</div>
  <div><strong>Author:</strong> ${book.author_name?.[0] ?? "Unknown"}</div>
  <div><strong>First published:</strong> ${book.first_publish_year ?? "Unknown"}</div>
`;
```

As seen in “infobox.js” line 28-32

By using different APIs through RESTful requests and a dynamic DOM, the website has been able to demonstrate how it’s possible to retrieve data from different sources and combine them into a coherent and data-driven narrative.

The data is parsed and selectively chosen to display relevant information regarding the website’s theme

Description of

Website design

The website is designed as a single page with a focus on a narrative, and data driven application, Goal 5, Target 5.3. The overall structure follows a -beginning-middle-end narrative, as described in the given section.

Implementation

The website has been implemented using HTML, CSS and JavaScripts: “Index.html”, “templatemo-style.css”, “infobox.js”.
HTML is used for the structural layout of my webpage, including containers for dynamic content like infoboxes, articles, books etc. Most of my website stems from a Template with a very long HTML code for the single-page structure. Most of the website is not hardcoded in HTML but instead populated programmatically with API, datasets and embeddings. A section has been “hardcoded” but mainly precoded by the template which is the section “Survivors”, I spent time selecting random photos and relevant stories related to my topic because I wanted to learn more about forced marriage and FGM. So this section gave me a lot of non-coding knowledge.

CSS is to control the layout, typography and visual consistency. The only thing I have changed here is the color of website, played a little bit with the sizes of the different sections - which took me a while.

JavaScripts is for retrieving data from external sources using DOM as mentioned in earlier sections.

Discussion

What have I learned about the problem in my “problem formulation” which was to

“*Raise awareness on forced marriage and female genital mutilation and suggest specific actions that people can take to achieve gender equality and empower all women and girls*”

And in addition to that build a data-driven web application with publicly available data. The application will then demonstrate how data can be transformed into meaningful information for users while raising awareness on forced marriage and FGM.

The project provided me with knowledge on the problem of gender inequality. But also gave important insights on the technical challenges that can occur when communicating sensitive social issues through data-driven applications. It has been learned that these harmful practices are widespread and unevenly distributed across regions and how they are deeply influenced by social and cultural factors.

Working with the datasets has given a highlight on how the data within this field is often incomplete, delayed of unevenly reported with a lot of missing values. This limitation did and does influence how the data can be visualized since there is a lot of missing data due to socio-economic factors. The raw data here is insufficient and must be contextualized through narrative, design and information. The information in some of the sections are harsh, but relevant. The images are happier and more neutral to make it “eatable” for the user. Presenting this information did require

choices regarding language and visual representation to ensure that the content is informed rather than sensationalized.

Conclusion

This project managed to implement and design a website while addressing gender inequality to empower women and girls. The focus was on eliminating harmful practices. The website integrated multiple web APIs and open-access datasets to demonstrate how raw data can become very meaningful information and knowledge with the help of visualizations, structure and design.
The project illustrates how websites can retrieve and display data to raise awareness, understanding on global issues. Future improvements could include more datasets, other interactive features. Personally, I would prefer to have a multi-page website.

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https://childmarriagedata.org/data-centre/

APIs
NYTimes

https://www.nytimes.com/topic/subject/female-genital-mutilation

Google Maps

OMBD (it interfered with the my Youtube video, so its not shown)
https://www.omdbapi.com/?apikey=a28801&i=tt1054580

Google Fonts – Open Sans.
https://fonts.google.com/

Youtube
https://www.youtube.com/watch?v=UKT1MX-RnmU

Open Library
https://openlibrary.org/works/OL1872814W/Wu%CC%88stenblume?edition=key%3A/books/OL22536552M

Appendices

Web page, javascript, css, dataset and images. All included in the attached "PrincessWebPage.zip"